DIGITAL COLLEGE LIBRARY | MIT

UML Diagram Report

Website overview, workflows, and conceptual system models

Project Overview

Digital College Library (MIT) is a campus collection discovery and circulation website. Readers can search by title, author, or ISBN; browse subjects; filter by availability; and inspect book, copy, and pickup-location details. Students can register or sign in, reserve an available campus copy for 24 hours, and use My Shelf. Librarians manage titles, copies, locations, checkouts, and returns. Admins manage accounts, roles, and access.

Diagrams Included

- DFD Level 0 context, Level 1 processes, and Level 2 category browsing
- UML Use Case Diagram for Visitor, Student, Librarian, and Admin
- UML Class Diagram for the proposed library domain
- UML Sequence Diagram for login success and invalid-credentials paths
- ERD for the proposed logical data model

Modeling note. The actual framework, backend, database, and storage implementation are unverified. Diagrams and data structures are conceptual. “Read” means finding and borrowing a campus copy; online e-book reading is not verified.

Project Overview and Problem Statement

Project overview

The public interface combines catalog discovery with copy-level campus inventory. Search accepts title, author, or ISBN. Subject browsing and availability choices narrow results. A detail view presents identifiers, copy counts, copy states, and pickup locations. The site also describes student holds and My Shelf, alongside separate librarian and administrator work areas.

Problem statement

Readers need to identify a relevant title, confirm that a physical campus copy is available, and find where to collect it. A title-only listing cannot show which individual copy is on shelf, checked out, or reserved. Library staff also need to maintain catalog records and item-level inventory, record checkout and return activity, and provide role-appropriate account access.

Objectives and Main Features

Objectives

- Make the campus catalog searchable by title, author, or ISBN.
- Let users browse subjects and filter results by copy availability.
- Show title metadata, copy states, counts, and pickup location together.
- Let students register, sign in, reserve an available copy for 24 hours, and use My Shelf.
- Give librarians catalog and circulation tools, and admins account and role controls.

Main features visible on the website

Feature	Visible behavior
Search	Search by book title, author, or ISBN.
Subject browsing	Choose a subject such as Computer science, Business, Literature, Science, Python, or Engineering mathematics.
Availability	Choose All availability, Available now, or Checked out.
Book and copy details	View author, ISBN, year, owned / available / borrowed / reserved counts, individual copy state, and library / floor / area / shelf / rack.
Account and hold	Register or sign in; reserve a specific available campus copy for 24 hours and view its pickup location.
My Shelf	Open the student area for personal loan tracking.
Staff operations	Librarians manage titles, copies, locations, checkouts, and returns. Admins manage accounts, roles, and access.

Target Users and Roles

Role	Primary responsibilities
Visitor	**Search the collection, browse subjects, filter by availability, and view book, copy, and pickup-location information.**
Student	**Use visitor discovery functions; register and sign in; reserve an available copy for 24 hours; use My Shelf; collect and borrow a campus book.**
Librarian	**Maintain titles, physical copies, and locations; record checkouts and returns.**
Admin	**Manage user accounts, assign roles, and control access to library functions.**

Visitor is treated as a guest role. Student, Librarian, and Admin are account roles in the conceptual model; the live authorization rules are not publicly documented.

Category Browsing and Filtering

The catalog combines a search field, a subject list, and an availability selector. A reader can begin with Everything or choose a subject, then filter the results to all availability, copies available now, or checked-out copies. Search accepts title, author, or ISBN. Result cards summarize subject, year, author, and available-copy count. Opening a title reveals ISBN and owned, available, borrowed, and reserved counts, with copy states and campus pickup details.

System workflow

- A Visitor or Student searches by title, author, or ISBN, or chooses a subject.
- The user selects an availability filter; matching titles and copy availability are shown.
- The user opens a title to review identifiers, counts, copy states, and pickup location.
- A Student registers or signs in, then reserves one available copy for 24 hours.
- The Student collects and borrows the campus copy; a Librarian records checkout and return activity.
- A Librarian maintains titles, copies, and locations. An Admin manages accounts, roles, and access.

Technology Stack and Conceptual Data Design

Technology stack evidence

The page displays a “Powered by Netlify” badge. This confirms a visible hosting or deployment brand only. The public interface does not establish the framework, server-side runtime, API design, database product, or storage provider; this report does not guess those details.

Conceptual architecture and storage

The proposed design separates catalog metadata from physical inventory, identity, reservations, and loans. These are logical records and stores for analysis, not verified deployed components. Availability is derived from individual copy state; title-level counts summarize its copies.

Conceptual records	Purpose and key relationships
Title / Author / Subject	**Describe catalog titles; title-to-author and title-to-subject links may be many-to-many.**
Copy / Location	**Represent each physical campus copy and its library, floor, area, shelf, and rack.**
Account / Role	**Represent signed-in users and Student, Librarian, or Admin permissions.**
Reservation	**Link one account to one selected copy and store creation and expiry times; the visible hold period is 24 hours.**
Loan	**Link an account to a physical copy and record checkout and return activity for My Shelf.**

Benefits, Limitations, and Future Scope

Benefits

- Search keys and subject browsing support different discovery paths.
- Availability and copy-level details help distinguish an available campus copy from a checked-out title.
- Shelf, rack, floor, and library details make campus pickup more direct.
- A 24-hour hold gives students a defined pickup window; role descriptions separate reader and staff tasks.

Limitations and verification boundaries

- Only the public website and its descriptions were reviewed; backend behavior and persistence are unverified.
- The visible interface describes campus copies; it does not verify online e-book reading.
- Loan length, eligibility, account recovery, retention, backups, accessibility conformance, and inventory synchronization are not specified or independently verified.

Future scope

- Document eligibility, checkout duration, return handling, and reservation expiry rules.
- Define persistent services for catalog, copy availability, accounts, reservations, and loans.
- Add auditable inventory and circulation workflows, conflict safeguards, accessibility review, and operational reporting.

Source. Digital College Library (MIT), public website, https://digital-libarys.netlify.app/ (accessed 27 October 2026). Website labels and functions are taken from the visible page; counts and live availability may change.

Data Flow Diagram (DFD) - Level 0 Context

One process exchanges catalog, account, reservation, circulation, and administration information with the four role groups.

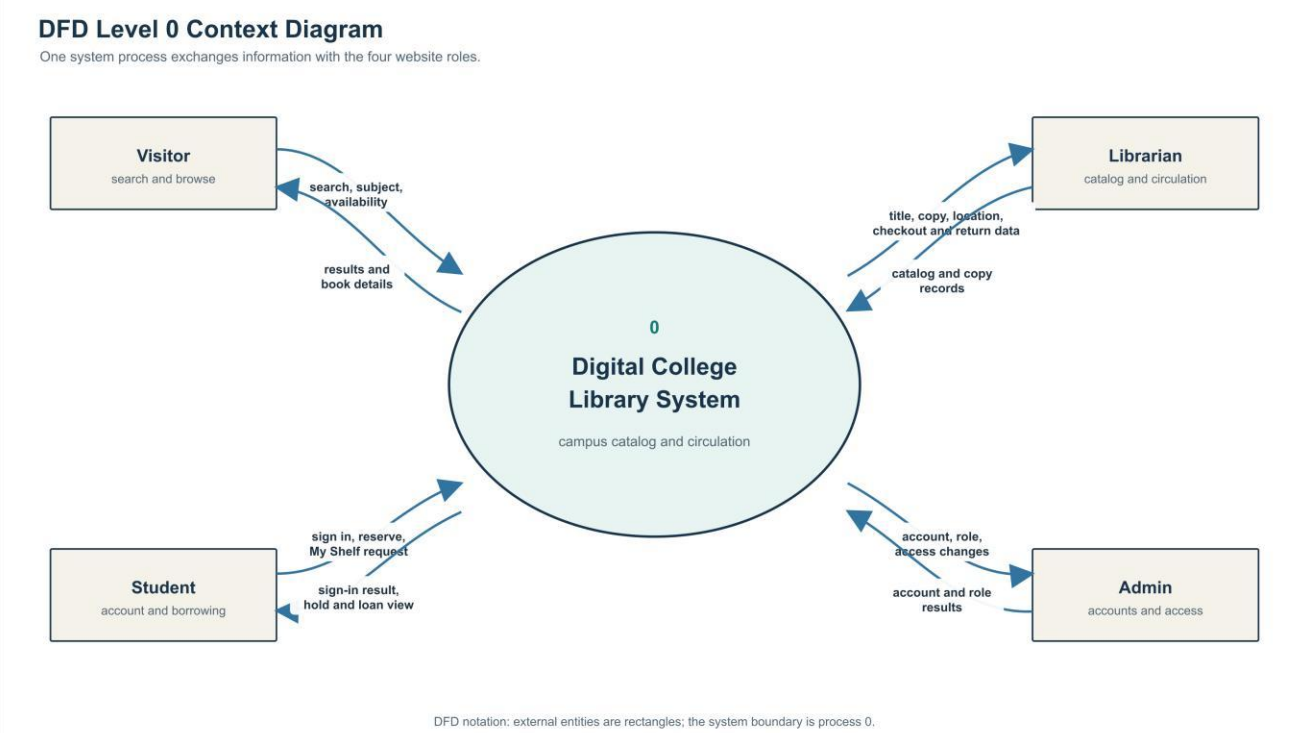


Figure 1. Context DFD for the Digital College Library system.

The context process accepts discovery, sign-in, hold, staff, and admin requests and returns catalog, copy, account, reservation, and circulation information.

Data Flow Diagram (DFD) - Level 1 Processes

The Level 1 view separates discovery, sign-in, reservations, catalog maintenance, circulation, and account administration.

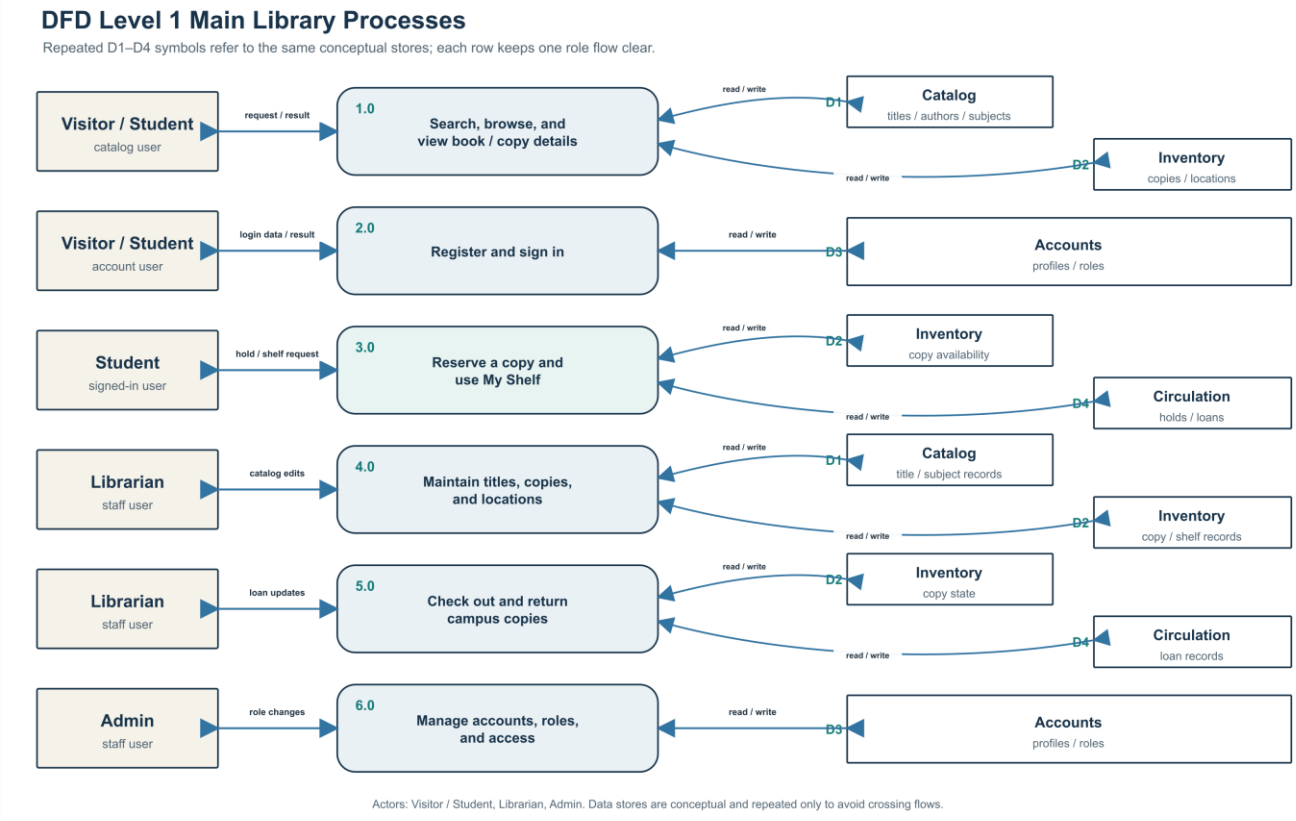


Figure 2. Level 1 DFD with main processes and conceptual data stores.

Repeated D1-D4 symbols refer to the same conceptual stores; processes and stores are not verified deployed components.

Data Flow Diagram (DFD) - Level 2 Category Browsing

A subject selection retrieves matching titles; copy state and location support availability filtering and campus catalog results.

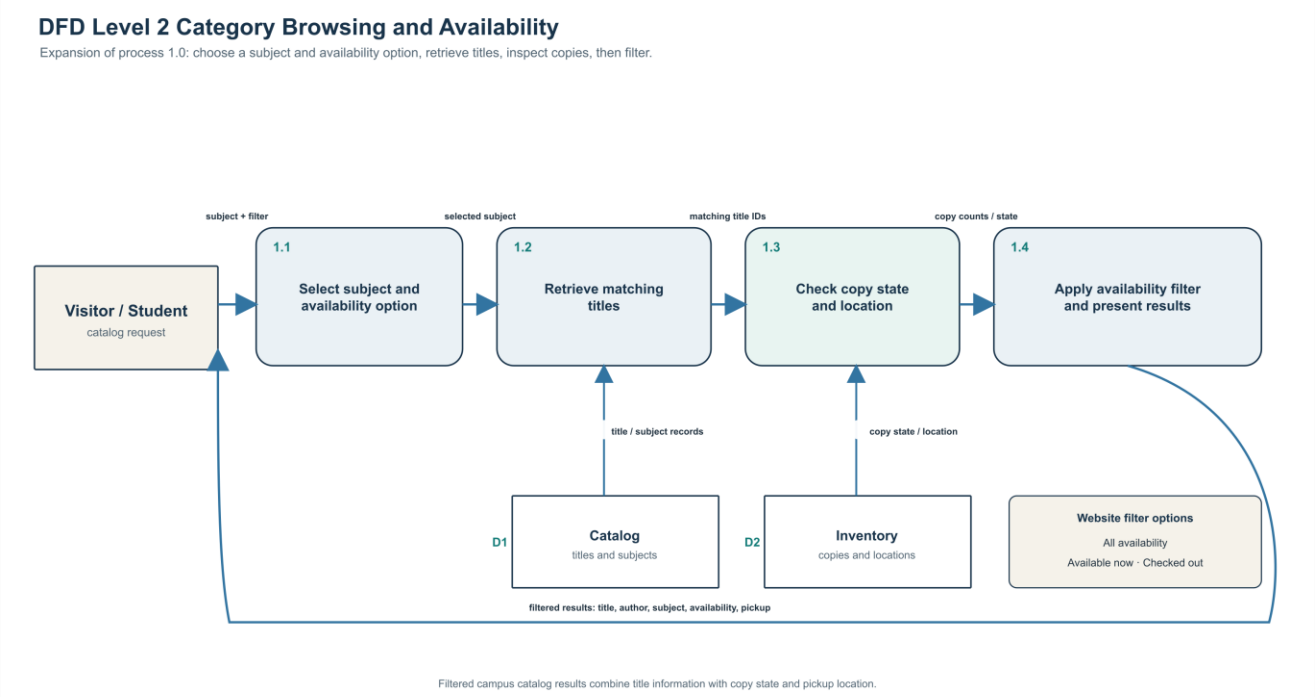


Figure 3. Level 2 DFD for category browsing and availability filtering.

This decomposition follows subject selection, title retrieval, copy-state lookup, availability filtering, and results presentation.

UML Use Case Diagram

The model maps public discovery, student account and hold functions, librarian catalog and circulation work, and admin account and access management.

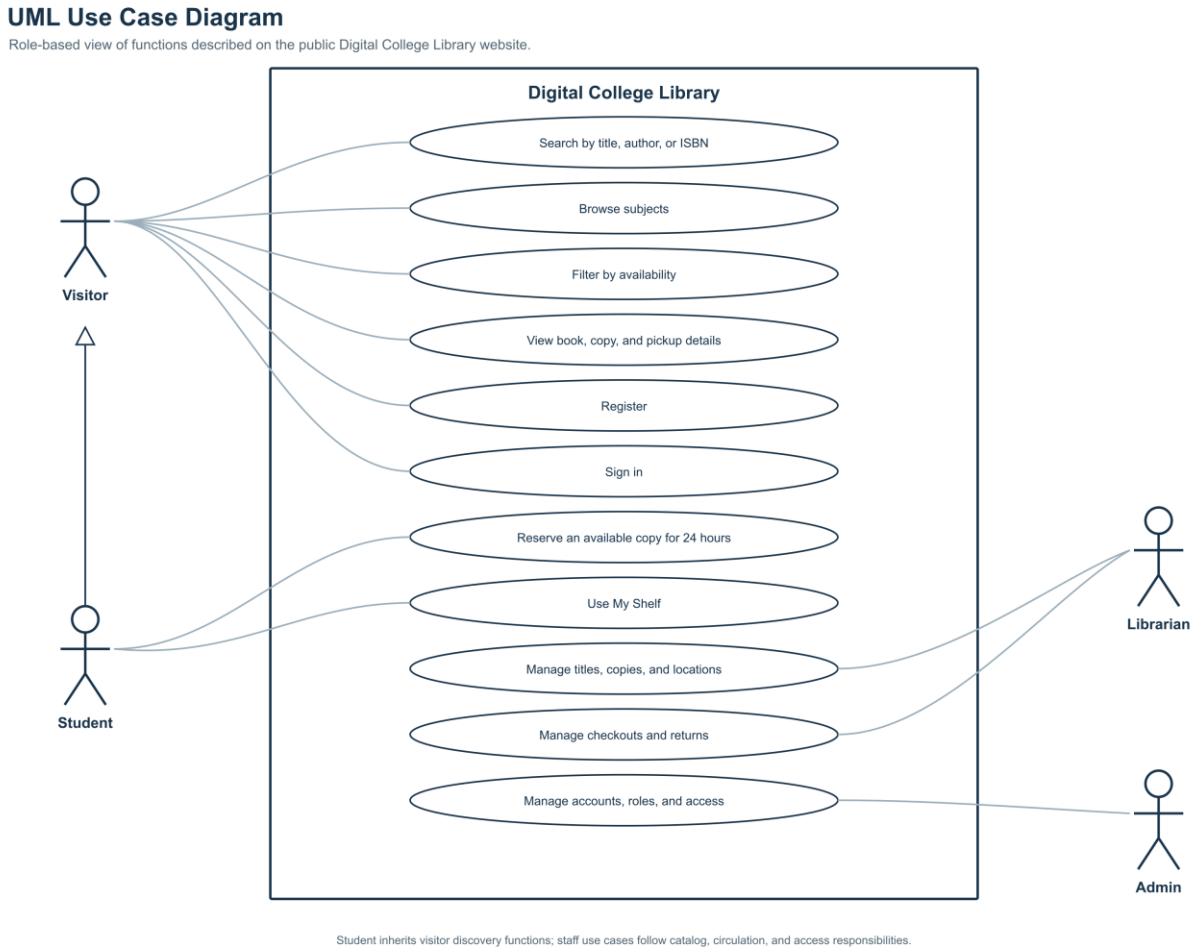


Figure 4. UML use case diagram for Visitor, Student, Librarian, and Admin.

The use case model reflects the user-facing role descriptions visible on the website.

UML Class Diagram

The conceptual model separates title metadata from copies, locations, reservations, loans, accounts, and roles.

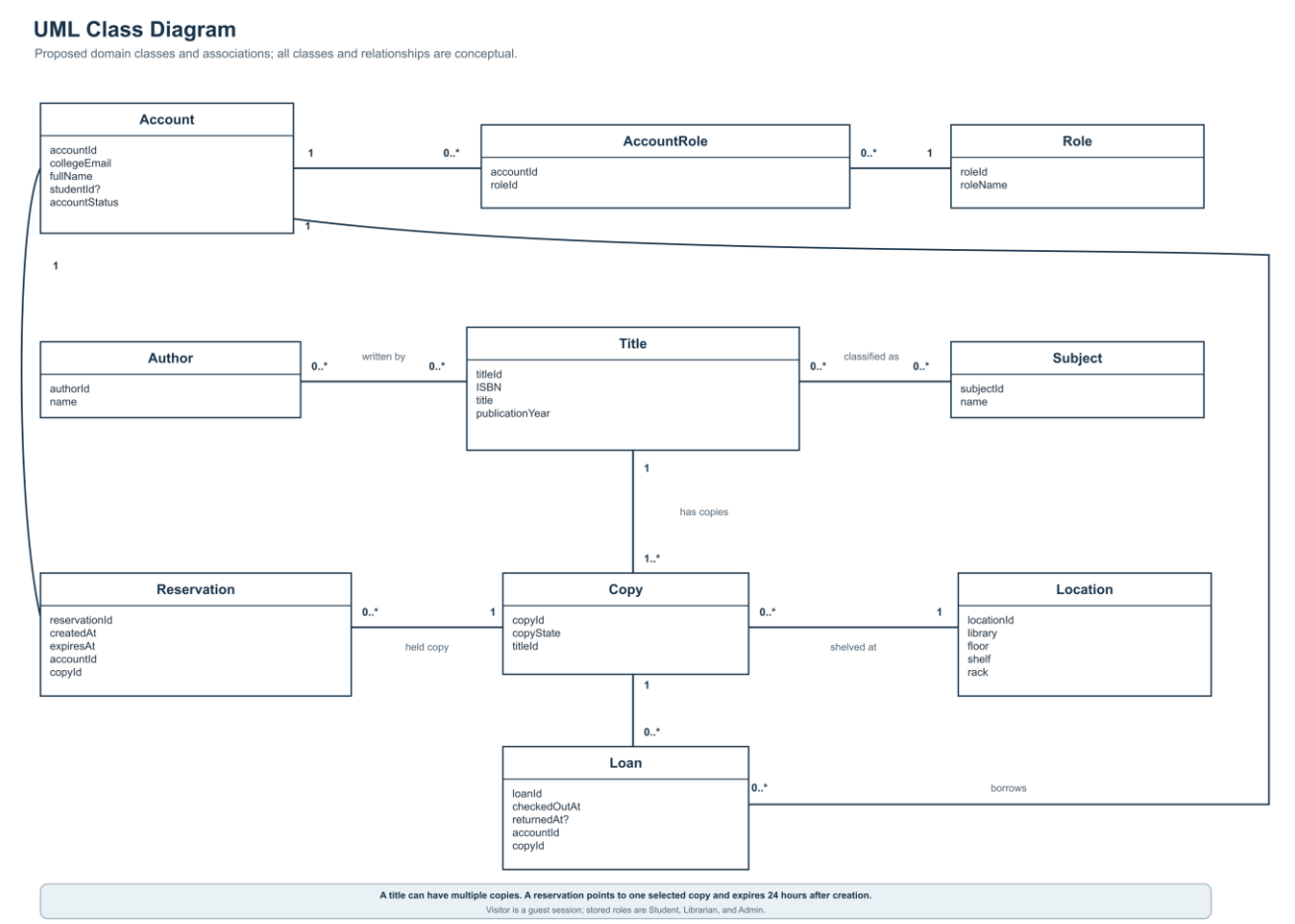


Figure 5. UML class diagram for the proposed library domain.

Class names, attributes, and relationships are proposed domain concepts, not verified website implementation details.

UML Sequence Diagram - User Login

The user submits a college email and password. Conceptual authentication returns a signed-in state or a retryable failure.

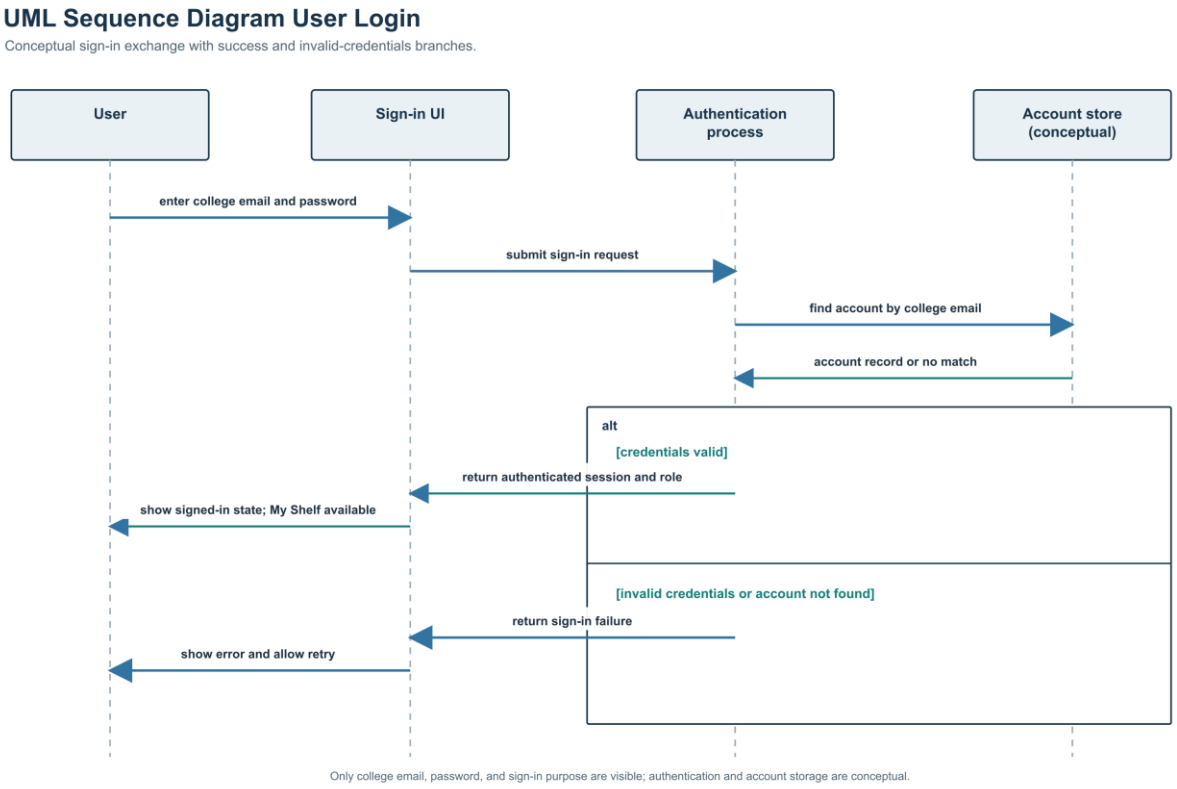


Figure 6. UML sequence diagram for successful and invalid-credential login paths.

Account data storage and authentication processing are conceptual; website implementation is not verified.

Entity Relationship Diagram (ERD)

The proposed model relates accounts, roles, catalog records, physical copies, locations, reservations, and loans.

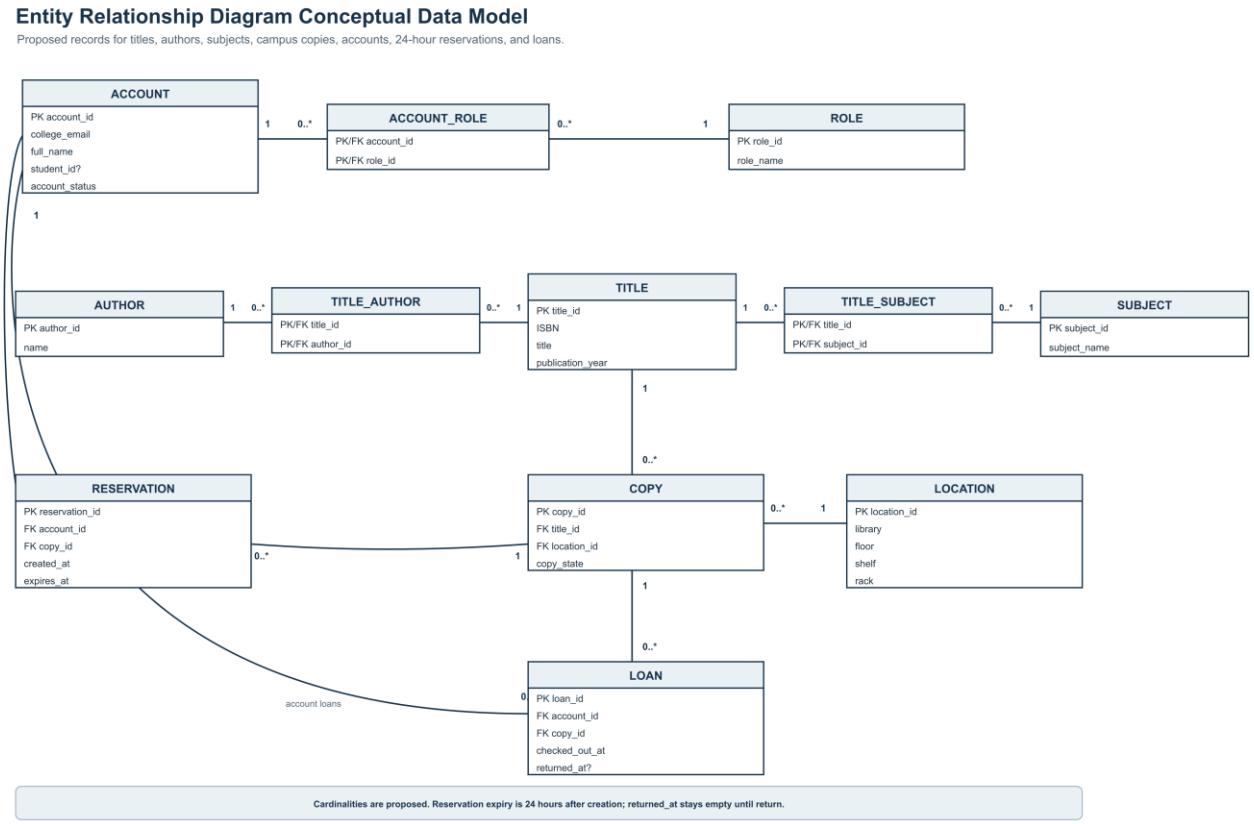


Figure 7. ERD for the proposed logical data model.

Primary and foreign keys show possible relationships; this is not a verified website schema.

Creating These Diagrams in Sparx Enterprise Architect

1. Open Enterprise Architect and open or create your Digital College Library project.

2. Create a package named “Digital College Library Diagrams” in the Browser panel.

3. Right-click the package and choose Add Diagram. Create one diagram for each of the five types.

4. For the Use Case, Class, and Sequence diagrams, select the matching UML diagram type and use the Toolbox elements.

5. For the DFD, use a suitable Data Flow/Flowchart notation available in your installed version. If DFD is not listed, use a suitable diagram and clearly label the DFD levels.

6. For the ER diagram, use the database/ERD notation available in your Sparx version; define primary keys, foreign keys, and relationship cardinalities.

7. Add labeled connectors, keep the layout readable, and export each diagram from Enterprise Architect as an image or PDF when required.

Modeling Assumptions

The diagrams are conceptual documentation based on the Digital College Library features described for this project. Verify that the actual implementation and Supabase schema use the same entities, fields, and authentication flow before final submission.